

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION

Federal State Autonomous Educational
Institution of Higher Education "Southern Federal University"
(SOUTHERN FEDERAL UNIVERSITY)

COURSE SYLLABUS FOR
Special topics in mathematics (Специальные разделы математики)

Подписано электронной подписью:
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I. GOALS AND OBJECTIVES OF MASTERING THE DISCIPLINE

Objectives of mastering the discipline:

- formation of a mathematical foundation for the analysis, modeling and research of processes in the field of information technology and artificial intelligence;
- development of systems thinking and an interdisciplinary approach to solving professional problems in the field of IT and AI;
- mastering modern mathematical apparatus for setting and solving promising problems of artificial intelligence;
- Formation of an understanding of basic mathematical methods and modeling for working with data uncertainty.

Objectives of mastering the discipline:

- study of the basic concepts and structures of special sections of mathematics;
- formation of ideas about mathematical modeling of processes and phenomena in information systems and artificial intelligence.
- Study mathematical approaches in optimization methods for training and tuning machine learning models.
- development of the ability to integrate mathematical methods into engineering solutions for the tasks of the IT industry and artificial intelligence.

II. THE PLACE OF THE DISCIPLINE IN THE STRUCTURE OF THE EDUCATIONAL PROGRAM

The discipline belongs to the compulsory part of the educational program and is included in the module B1.M3.ED1 of elective disciplines.

This discipline is based on the basic knowledge, skills and abilities formed when obtaining the previous level of education and the course "Higher Mathematics".

Knowledge, skills and abilities formed by this discipline will be required when mastering the following elements of the educational program:

- "Generative Artificial Intelligence and Machine Learning"
- "Artificial Intelligence and Data Analysis";
- "Architecture, Technologies and Tools for the Development of AI-Enabled Software";
- "Algorithms, Technologies and Tools for Image and Video Sequence Processing"
- "Machine Learning in Computer Vision"
- "Optimization methods";
- "Numerical methods";
- "Mathematical Methods of Image Recognition";
- "Theory of Decision-Making";
- "Business Analytics";
- "Cryptographic methods of information protection";
- "Evolutionary modeling".

III. REQUIREMENTS FOR THE RESULTS OF MASTERING THE DISCIPLINE

The development of the discipline is aimed at the formation of the following competencies in accordance with the educational standard and educational program:

List of planned learning outcomes in the discipline correlated with indicators of competence achievement

Competence	Indicators of Competency Achievement	Learning outcomes
CPC-1 – Able to understand the natural science essence of problems that arise in the course of professional activity, apply interdisciplinary and general scientific knowledge and methods to solve problems of analysis, modeling and research of objects of professional activity	CPC-1.1 - Solves standard professional problems using natural science and general engineering knowledge, methods of mathematical analysis and modeling	Must know: - basic mathematical methods for describing objects; - determination and methods for calculating multiple, curvilinear and surface integrals; - basic methods for solving ordinary differential equations; - the concept of convergence and types of numerical and functional series, including Fourier series in the analysis of periodic functions; - basic concepts and methods of the theory of the function of a complex variable and operational calculus. Must be able to: - analyze functions: find extremes, analyze charts; - apply mathematical concepts to solving practical problems; - Calculate integrals of various types (multiple, curvilinear, surface); - apply various methods to solve differential equations. Must be proficient in: - methods of solving problems using algorithms and methods of mathematical analysis; - skills in working with data and their visualization in the context of mathematical models; - methods of modeling processes based on differential equations.
	CPC-1.2 - Conducts theoretical and experimental studies of objects of professional activity	
SC 3. MF-3 Capable of applying modern optimization techniques to train machine learning models, tune hyperparameters, and solve artificial intelligence problems	SC 3.1 MF-3.1 Applies optimization techniques to the development and research of training algorithms	Must know: - basic mathematical tools for training machine learning models and tuning hyperparameters; - Mathematical foundations of convergence and stability of optimization algorithms. Must be able to: - perform analytical solutions to the simplest optimization problems; - Interpret the mathematical results of optimization in the context of machine learning and artificial intelligence problems. Must be proficient in: - skills of formalizing AI tasks in mathematical form; - the methodology for choosing an approach to optimization depending on the properties of the objective function; - a mathematical apparatus for assessing the stability and correctness of the solutions found.

IV. CONTENT AND STRUCTURE OF THE DISCIPLINE

The complexity of the discipline is 5 credits, 180 hours,
including 1 credit, 36 hours for exams

Form of intermediate attestation: exam (3rd semester)

4.1. The content of the discipline, structured by the topics of classes, semester 3

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
Module 1. Functions of multiple variables										
1	Topic 1. Substitution of variables in a double integral, polar coordinate system 1.1. General view of variable substitution in a double integral 1.2. Polar coordinate system for a double integral	2	-	-	-	-	Informational lecture	-	-	-
2	General view of variable substitution in a double integral	-	2	-	-	2	Seminar	-	-	-
3	Substitution of variables in a double integral, polar coordinate system	-	2	-	-	2	Seminar	-	-	-
4	Topic 2. The Triple Integral, Its Properties and Computation 2.1. Problems leading to the concept of the triple integral; Multidimensional data processing 2.2. Definition of the Triple Integral 2.3. Properties of the triple integral 2.4. Reduction of the triple integral to the repeated one 2.5. Substitution of variables in the triple integral 2.6. Cylindrical and spherical coordinate systems	2	-	-	-	-	Informational lecture	-	-	-
5	Calculation of the triple integral, application of its properties	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
6	Topic 3. Curvilinear and surface integrals of the 1st kind 3.1. Determination of the curvilinear integral of the first kind; lengths, areas, and multidimensional sums 3.2. Properties of the curvilinear integral of the 1st kind 3.3. Reduction of the curvilinear integral of the 1st kind to a definite one 3.4. Surface integral of the 1st kind, properties 3.5. Reduction of the surface integral of the 1st kind to a double integral	2	-	-	-	-	Informational lecture	-	-	-
7	Substitution of variables in the triple integral. Calculation of triple integrals in a cylindrical coordinate system	-	2	-	-	2	Seminar	-	-	-
8	Calculation of curvilinear and surface integrals of the 1st kind. Different ways to define a contour.	-	2	-	-	2	Seminar	-	-	-
9	Topic 4. Curvilinear and surface integrals of the 2nd kind. Formulas of Green, Ostrogradsky, Stokes 4.1. Scalar and vector fields; Processing of data streams 4.2. Curvilinear integrals of the 2nd kind 4.3. Surface integral of the 2nd kind 4.4. Green's formula 4.5. Ostrogradsky's formula 4.6. The Stokes formula	2	-	-	-	-	Informational lecture	-	-	-
10	Calculation of curvilinear and surface integrals of the 2nd kind. Formulas of Green, Ostrogradsky, Stokes.	-	2	-	-	2	Seminar	-	-	-
11	Topic 5. Scalar and vector field operations 5.1. Function Gradient 5.2. Divergence 5.3. Rotor 5.4. Operator recording of formulas. Gradient, divergence, and rotor operations 5.5. Directions of the greatest change in loss functions, analysis of local changes and flows of characteristics	2	-	-	-	-	Informational lecture	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
12	Calculating Gradient, Rotor, and Divergence	-	2	-	-	2	Seminar	-	-	-
13	Integration of FNP. Lesson with a test.	-	2	-	-	2	Seminar	Test No1	Counter. Work	20
Module 2. Rows. TAC										
14	Topic 6. Numerical series. A prerequisite for convergence. Signs of comparison 6.1. Definition of the numerical series, the sum of the series, the necessary feature of convergence 6.2. Signs of comparing the convergence of a numerical series 6.3. Sign of d'Alembert convergence 6.4. Cauchy Convergence Sign, Cauchy Radical Sign 6.5. Integral Feature of Convergence of a Numerical Series 6.6. Numerical series to substantiate the correctness and stability of machine learning methods	2	-	-	-	-	Informational lecture	-	-	-
15	Finding the limit of the numerical series. A prerequisite for convergence. Signs of comparison	-	2	-	-	2	Seminar	-	-	-
16	Topic 7. Alternating series. 7.1. Absolute and conditional convergence of alternating series; Evaluation of alternating components 7.2. Alternating series, Leibniz sign 7.3. Abel's sign 7.4. Assessing the Impact of Noise and Oscillations – Wavelets and Filters	2	-	-	-	-	Informational lecture	-	-	-
17	Signs of convergence of sign-positive numerical series	-	2	-	-	2	Seminar	-	-	-
18	Determination of the convergence of alternating series. Alternating series, Leibniz's sign. Abel's sign	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
19	Topic 8. Power series. 8.1. General properties of functional series 8.2. Power series, Abel's theorem, radius of convergence 8.3. Decomposition of functions into power series 8.4. Complex power series in the approximation of complex models	2	-	-	-	-	Informational lecture	-	-	-
20	Finding the radius and convergence region of a power series.	-	2	-	-	2	Seminar	-	-	-
21	Topic 9. Fourier trigonometric series 9.1. Orthogonality of the trigonometric system. Fourier series, Fourier coefficients 9.2. Dirichlet's theorem. Fourier series expansion of a periodic function 9.3. Even and odd functions, decomposition into a series 9.4. Fourier series in complex form 9.5. Fourier transform - spectral analysis and decomposition of signals on a basis	2	-	-	-	-	Informational lecture	-	-	-
22	Decomposition of a periodic function into a Fourier series.	-	2	-	-	2	Seminar	-	-	-
23	Decomposition of a periodic function into a Fourier series. Even and odd functions.	-	2	-	-	2	Seminar	-	-	-
24	Topic 10. Problems leading to differential equations (DU), first-order linear differential equations (LDU) 10.1. Tasks leading to remote control - modeling of process dynamics 10.2. Definition of the Differential Equation, Cauchy Problem 10.3. Remote controls with and reducible variables 10.4. First-order LDL, method of variation of an arbitrary constant 10.5. First order LDL, Bernoulli method 10.6. Bernoulli's equation	2	-	-	-	-	Informational lecture	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
25	Tasks leading to remote control, general concepts. DUs with and reducible variables	-	2	-	-	2	Seminar	-	-	-
26	Topic 11. Equation in complete differentials and order reduction in ordinary differential equations (TACs) 11.1. Equation in complete differentials 11.2. Lowering the order in the trust management 11.3. Evolution of Neural ODEs	2	-	-	-	-	Informational lecture	-	-	-
27	Application of the Bernoulli equation.	-	2	-	-	2	Seminar	-	-	-
28	Equations in complete differentials and methods for lowering the order of the TAC	-	2	-	-	2	Seminar	-	-	-
29	Topic 12. Linear homogeneous differential equations (LODU) of higher orders 12.1. General properties of the LODU 12.2. LODU with constant odds 12.3. TAC Order Reductions for Analysis of Complex Dynamical Systems, Including Neural TACs and Continuous Flows	2	-	-	-	-	Informational lecture	-	-	-
30	Decision of the LODU of the highest orders. DD of the 2nd order. Classification of equations by solution methods	-	2	-	-	2	Seminar	-	-	-
31	Topic 13. Linear inhomogeneous differential equations (LNDU) of higher orders 13.1. LNDU with constant coefficients, special view of the right side 13.2. LNDU with constant coefficients, method of variation of arbitrary constants 13.3. Systems of differential equations in multidimensional systems in forecasting problems	2	-	-	-	-	Informational lecture	-	-	-
32	Rows and ODE. Lesson with test work	-	2	-	-	2	Seminar	Test No2	Counter. Work	20
33	Decision of the LNDU of the highest orders. LNDU of the highest orders with a special right part	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
34	Topic 14. Development of the concept of numbers, complex numbers, geometric meaning of operations with them 14.1 Development of the concept of number, numerical sets 14.2 Complex numbers, operations with them 14.3 Complex numbers as vectors on a plane 14.4. Trigonometric form of notation 14.5. The Geometric Meaning of Multiplication and Division of Complex Numbers 14.6. Indicative Form of Complex Number Notation 14.7. Exponentiation of Complex Numbers 14.8. Extracting the nth Degree Root from Complex Numbers	2	-	-	-	-	Informational lecture	-	-	-
35	Complex numbers, operations on them.	-	2	-	-	2	Seminar	-	-	-
36	Topic 15. Complex Variable Function (FCP), a derivative of FCP 15.1. FKP. The real and imaginary part of the function 15.2. Limit and continuity of the function. Basic Elementary Functions 15.3. Derivative and differential of a function of a complex variable. Computational accuracy for complex data 15.4. Cauchy-Riemann conditions, analytic functions 15.5. Geometric meaning of the derived function of a complex variable 15.5. Application of FCP in the analysis of the stability of filters and systems	2	-	-	-	-	Informational lecture	-	-	-
37	Solving equations with complex roots	-	2	-	-	2	Seminar	-	-	-
38	Derivative and differential of a function of a complex variable. Cauchy-Riemann conditions	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
39	Topic 16. Integrating complex variable functions 16.1. Integral of the function of a complex variable. Its properties and applications in data analysis and processing 16.2. Cauchy's Theorem for Singly Connected and Many-Connected Domains 16.3. Independence of the integral from the path of integration 16.4. Cauchy's Integral Formula 16.5. Infinite Differentiability of a Function Analytic in the Domain 16.6. Taylor series. Decomposition of the basic elementary functions of a complex variable into a Taylor series 16.7. Laurent's series, its properties; Application in complex spectral analysis	2	-	-	-	-	Informational lecture	-	-	-
40	Integration of the FKP. Application of the Cauchy formula	-	2	-	-	2	Seminar	-	-	-
41	Topic 17. Isolated singular points and function deduction 17.1. Isolated singular points, their classification 17.2. Deduction, the Basic Theorem of the Theory of Deductions 17.3. Calculation of the deduction 17.4. Use of the FKP deduction 17.5. Calculating Integrals Using Subtractions	2	-	-	-	-	Informational lecture	-	-	-
42	Decomposition of the basic elementary functions of a complex variable into Taylor and Laurent series	-	2	-	-	2	Seminar	-	-	-
43	Classification of singular points. Calculating Integrals Using Subtractions	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
44	Topic 18. Laplace transform and its properties 18.1. Laplace transform; Application in the processing of dynamic and time series 18.2. Properties of the Laplace Transform 18.3. Image Table 18.4. Solving the Cauchy Problem for ODE by the Operational Method - Quick Solutions for Dynamical Systems, Multivariate Process Modeling, and Feature Prediction 18.5. Solution of ODE systems by the operational method	2	-	-	-	-	Informational lecture	-	-	-
45	Elements of TFCP and OI. Lesson with test work	-	2	-	-	2	Seminar	Test No3	Counter. Work	20
Intermediate certification		-	—	—		36	Exam	Copies. question. and tickets for semester 2	Exam	40
Total Hours: 180		36	54	-		90		—		100

4.2. Plan of extracurricular independent work

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1. Functions of multiple variables						
1	Topic 1. Substitution of variables in a double integral, polar coordinate system	3	6	- Study and repetition of the material of lectures and practical classes	4	[2], [7], [8], [11], additional [16] – [19]
2	Topic 2. The Triple Integral, Its Properties and Computation	3	7	- Study and repetition of the material of lectures and practical classes	2	[2], [7], [8], [11], additional [16] – [19]
3	Topic 3. Curvilinear and surface integrals of the 1st kind	3	8	- Study and repetition of the material of lectures and practical classes	4	[2], [7], [8], [11], additional [16] – [19]
4	Topic 4. Curvilinear and surface integrals of the 2nd kind. Formulas of Green, Ostrogradsky, Stokes	3	8	- Study and repetition of the material of lectures and practical classes - Preparation for the test	2	[2], [7], [8], [11], additional [16] – [19]
5	Topic 5. Scalar and vector field operations	3	9	- Study and repetition of the material of lectures and practical classes	4	[2], [7] - [9], [11], additional [16] - [19]
Module 2. Rows. TAC						
6	Topic 6. Numerical series. A prerequisite for convergence. Signs of comparison	3	10	- Study and repetition of the material of lectures and practical classes	2	[2], [7] - [9], [11], additional [16] - [19]
7	Topic 7. Alternating series	3	10	– - Study and repetition of the material of lectures and practical classes	4	[2], [7] - [9], [11], additional [16] - [19]
8	Topic 8. Power series	3	11	– - Study and repetition of the material of lectures and practical classes	2	[2], [7] - [9], [11], additional [16] - [19]
9	Topic 9. Fourier trigonometric series	3	12	– - Study and repetition of the material of lectures and practical classes	4	[2], [4], [7] - [9], [11], additional [15] - [19]
10	Topic 10. Problems leading to differential equations (DU), Linear differential equations of the first order	3	12	- Study and repetition of the material of lectures and practical classes	2	[2] - [4], [7] - [9], [11], additional [15] - [19]
11	Topic 11. Equation in complete differentials and order reduction in ordinary differential equations (TACs)	3	13	– - Study and repetition of the material of lectures and practical classes	4	[2] - [4], [7] - [9], [11], additional [15] - [19]
12	Topic 12. Linear homogeneous differential equations (LODU) of higher orders	3	14	– - Study and repetition of the material of lectures and practical classes	2	[2], [4], [7] - [9], [11], additional [15] - [19]
13	Topic 13. Linear inhomogeneous differential equations (LNDU) of higher orders	3	14	- Study and repetition of the material of lectures and practical classes - preparation for the test	4	[2],[3], [4], [7] - [9], [11], additional [15] - [19]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
14	Topic 14. Development of the concept of numbers, complex numbers, geometric meaning of operations with them	3	15	– - Study and repetition of the material of lectures and practical classes	2	[23], [4], [7] - [9], [11], additional [15] - [19]
15	Topic 15. Function of a complex variable, a derivative of the FCP	3	16	- Study and repetition of the material of lectures and practical classes	4	[2], [4], [7] - [9], [11], additional [15] - [19]
16	Topic 16. Integrating complex variable functions	3	16	- Study and repetition of the material of lectures and practical classes	2	[2], [4], [7] - [9], [11], additional [15] - [19]
17	Topic 17. Isolated singular points and function deduction	3	17	- Study and repetition of the material of lectures and practical classes - preparation for the test	4	[2], [4], [7] - [9], [11], additional [12] - [17], [22]
18	Topic 18. Laplace transform and its properties	3	18	- Study and repetition of the material of lectures and practical classes	2	[2], [4], [7] - [9], [11], additional [12] - [17], [22]
The total labor intensity of independent work in the discipline					54	-
Exam preparation					36	-
The total labor intensity of independent work in the discipline					90	-

V. EDUCATIONAL AND METHODOLOGICAL SUPPORT OF THE DISCIPLINE

5.1. References

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12. Melekhov A.P., Prozorov O.A., Revina S.V., Sazonov L.I. Textbook "Construction of Phase Portraits of Dynamic Systems. Theory and Software Implementation" on the course "Differential

- Equations" for the bachelor's degree program in the direction of "Applied Mathematics and Informatics" SFedU Publishing House, 2008: <https://hub.sfedu.ru/storage/1/869632/ac21e93f-56e7-40df-a9cd-fb51a7cdd1b5/>
13. Revina S.V., Sazonov L.I., Tsyvenkova O.A. Equations of Mathematical Physics. Problems and Solutions. SFedU Publishing House, 2019: <https://hub.sfedu.ru/storage/1/869593/6666cde0-flc7-45fd-979f-9bd018a03fc4/>
 14. Erusalimsky Y.M., Malonek G.R. This "simple" and "beautiful" mathematics. SFedU Publ., 2023: <https://hub.sfedu.ru/repository/material/801319480/>
 15. Shabarshina I. S., Korokhova E. V., Korokhov V. V. Fundamentals of Computer Mathematics. Problems of System Analysis and Control: Textbook / I. S. Shabarshina, E. V. Korokhova, V. V. Korokhov ; Southern Federal University. – Rostov-on-Don ; Taganrog : Publishing House of the Southern Federal University, 2019. – 142 p.: <https://hub.sfedu.ru/storage/1/819415/d717d4ca-084a-4c9a-b9a2-beaf18fe496b/>

5.2. Further reading

16. Aleksandriysky B.I. Textbook on the course "Differential equations of mathematical physics" [Text]: [for all specialties] / TRTU, EHF, Kaf. VM - Taganrog: TRTU Publ., 2005. - 71 p. (97 copies)
17. Collection of tasks for typical calculations and control works in mathematical disciplines [Text]. Part 1 / Editor-in-Chief A.I. Sukhinov; TRTU, EHF, Kaf. VM - Taganrog: TRTU Publ., 2006. - 545 p. (115 copies)
18. Smirnov V.I. Course of Higher Mathematics / V.I. Smirnov - Moscow: State Publishing House of Phys.-Math. Lit., 1959. - 656 p. <http://biblioclub.ru/index.php?page=book&id=257406>.
19. Erusalimsky Y.M., Malonek G.R. This simple and beautiful mathematics. - 2nd ed., ispr. Rostov-on-Don: Southern Federal University (SFU), 2022. — 260 p. — ISBN 9785927541553: <https://hub.sfedu.ru/repository/material/801319480/>.
20. Lecture notes on the course "Higher Mathematics2 [Text]. Part 1 / TRTU, EHF, Kaf. VM - Taganrog: TRTU Publ., 1999. - 388 p. (164 copies).
21. A.A. Afonin [i dr.] Metodicheskie ukazaniya k otdelu "Linear algebra" kursa "Vyshey matematiki" [Text]: for students. all special. EHF / TRTU, EHF, Kaf. VM: - Taganrog: TRTU Publ., 1998. - 41 p. (292 copies).
22. Lyapunova I.A., Kudrinskaya T.V. Solution of Equations of the 4th Degree: Practicum / I.A. Lyapunova, T.V. Kudrinskaya; Southern Federal University. – Rostov-on-Don, 2021. – 40 p.: <https://hub.sfedu.ru/storage/1/1297694/6644e448-0b0c-4ee7-a342-9ce79dd5b58b/>.
23. Lisyak V.V. Mathematical Foundations of Computer Graphics: Transformations, Projections, Surfaces: Textbook / V.V. Lisyak ; Southern Federal University. – Rostov-on-Don ; Taganrog : Publishing House of the Southern Federal University, 2020. – 103 p.: <https://hub.sfedu.ru/storage/1/1272904/b14cecf-d6c0b-4201-98d5-5c2b3b620003/>.
24. Galustov G.G., Sedov A.V. Mathematical Modeling and Forecasting of Processes in Technical Systems: Textbook / Galustov G.G., Sedov A.V. ; Southern Federal University. – Rostov-on-Don ; Taganrog: Publishing House of the Southern Federal University, 2016. – 107 p.: <https://hub.sfedu.ru/storage/1/914247/eed4b31c-9d9c-4b9b-b22d-740a3c067931/>.
25. Matveeva L.G. Economic and Mathematical Methods and Models in Innovation Management: Textbook / L.G. Matveeva ; Southern Federal University. – Rostov-on-Don ; Taganrog : Publishing House of the Southern Federal University, 2018. – 204 p.: <https://hub.sfedu.ru/storage/1/756900/6bfefaec-5c28-4aba-8bdf-49877e2b48d1/>.
26. Ivanova O.A., Y.S. Nalbandyan. Higher Mathematics. Practicum: Educational and Methodological Manual / O.A. Ivanova, Y. S. Nalbandyan; Southern Federal University. – Rostov-on-Don; Taganrog, 2020. – 126 p.: <https://hub.sfedu.ru/storage/1/1282226/bbcea9d3-3a51-4b97-8ce1-05d17f3b1e5e/>.
27. Shabarshina I.S., Korokhova E.V., Korokhov V.V. Fundamentals of Computer Mathematics. Problems of System Analysis and Control: Textbook / I. S. Shabarshina, E. V. Korokhova, V. V.

- Korokhov ; Southern Federal University. – Rostov-on-Don ; Taganrog : Publishing House of the Southern Federal University, 2019. – 142 p.: <https://hub.sfedu.ru/storage/1/819415/d717d4ca-084a-4c9a-b9a2-beaf18fe496b/>.
28. Chernyavskaya I.A. Analytical Geometry in Examples and Tasks: Textbook / I. A. Chernyavskaya; Southern Federal University. – Rostov-on-Don: Publishing House of the Southern Federal University, 2016. – 66 p.: <https://hub.sfedu.ru/storage/1/749288/afc688e5-afd1-4fb8-95d7-2e993ffdf0ae/>.

5.3. Periodicals

1. Proceedings of Higher Educational Institutions. Mathematics (<https://kpfu.ru/science/nauchnye-izdaniya/ivrm>)
2. Mathematical Collection (http://www.mathnet.ru/php/journal.phtml?jrnid=sm&option_lang=rus)
3. VMF. Series 1. Mathematics, mechanics (http://www.mathnet.ru/php/journal.phtml?jrnid=vmumm&option_lang=rus)

5.4. List of Internet resources

1. <http://ictis-lms.sfedu.ru/>. E-learning system of ICTIB SFedU
2. <https://hub.sfedu.ru/>. SFedU Electronic Resources Portal

5.5. Scientific publications at A* level conferences, articles published in level 1 journals of the White List of scientific journals

1. Vasin A. A., Davidson M. R., Novikova N. M. Mathematical models of energy network systems. Series 15: Computational Mathematics and Cybernetics. – 2024. – № 4. – P. 60-72. – DOI 10.55959/MSU/0137-0782-15-2024-47-4-60-72. – EDN KXFMKN.
2. Bikchentaev A.M. Block Design Operator in the Algebra of Measurable Operators. Proceedings of Higher Educational Institutions. Mathematics. 2023; (10):77-82. <https://doi.org/10.26907/0021-3446-2023-10-77-82>
3. O. I. Egorushkin, I. V. Kolbasina, K. V. Safonov, "On the solution of a general algebraic equation by power series and application in the theory of formal grammars", PDM, 2023, No 60, 106–113

VI. MATERIAL AND TECHNICAL SUPPORT OF THE DISCIPLINE

The following premises, equipment and software are used in the implementation of the discipline:

No p/n	Type of classroom lesson	Recommended audience	
		Type	Equipment
1	Lectures	lecture-type auditorium	– interactive whiteboard - 1 pc., teacher's computer - 1 pc.; – Microsoft Windows, Microsoft Office PowerPoint
2	Practical exercises	Seminar-type auditorium	– interactive whiteboard - 1 pc., teacher's computer - 1 pc.; – Microsoft Windows, Microsoft Office PowerPoint Microsoft Windows, Microsoft Office PowerPoint
3	Independent work	Audience of individual and group consultations, intermediate certification	– interactive whiteboard - 1 pc., teacher's computer - 1 pc.; Microsoft Windows, Microsoft Office PowerPoint Microsoft Windows, Microsoft Office PowerPoint

VII. METHODOICAL INSTRUCTIONS FOR STUDENTS ON MASTERING THE DISCIPLINE

7.1. General Provisions

It is recommended to familiarize yourself with the content of the course according to the work program of the discipline.

Write out (download) from the corresponding work program:

- list of recommended literature;
- the names of the lecture sections of the course;
- topics of practical classes;
- theoretical questions for the test and exam.

Students are recommended, in accordance with the schedule of lectures and practical classes in this discipline, to plan days of the week and hours for independent work, which will include preparation for lectures, practical classes, as well as preparation for intermediate control and exams.

7.2. Preparation for lectures (theoretical course)

Recommendations:

- before the next lecture, it is necessary to review the material of the previous lecture according to your notes;
- get acquainted with the content of the next lecture on the main sources of literature in accordance with the work program of the discipline;
- In case of difficulties in perceiving the material, it is necessary to turn to the main literary sources, lectures, to the lecturer - according to the schedule of his consultations, to the teacher - in practical classes.

7.3. Preparation for practical classes

Practical classes are aimed at consolidating theoretical knowledge, developing and forming practical skills and abilities in the course.

Recommendations:

- when preparing for a practical lesson, it is necessary to be guided by the content of the thematic plan in the work program of the discipline, i.e. to know the basic concepts, definitions, terminology, axioms and theorems;
- when preparing for the relevant practical lesson, it is necessary to look through the examples considered at the lecture from the lecture notes.

7.4. Preparation and implementation of the test

In accordance with the work program of the discipline, the study of the relevant sections ends with the completion and submission of the test work. The student should start performing the test work only after studying the theoretical material corresponding to the topic from the lecture notes or textbook, and analyzing the examples given. Options for tests are given in the fund of assessment tools (FOS) of the discipline.

Recommendations for implementation:

- the student completes the version of the test provided by the teacher;
- the test is carried out on notebook sheets (up to 2-4 pages);
- when solving problems, it is necessary to justify each stage of the solution, to give the necessary formulas;
- calculations should be sufficiently detailed, arranged in a strict order;
- the solution of each problem should be brought to the answer required in the condition;
- the solution of problems of geometric content should be accompanied by a drawing or sketch, made accurately, indicating the axes of coordinates and the unit of measurement;
- Explanations for the problems should correspond to the designations given in the figures.

Tests must be done independently. Undone work deprives the student of the opportunity to check the degree of his preparedness on the topic under study. If the teacher determines that the work is not independent, namely, the student will not be able to answer the teacher's questions on solving the tasks of the test, then it will be evaluated for the minimum number of points.

Having received a checked work from the teacher, the student must independently work on mistakes. In case of non-credit for the work, the student can fulfill all the requirements as soon as possible and submit the work for re-checking, attaching the originally completed work. Retaking the test is carried out as part of retakes conducted by the teacher. In cases where a student encounters difficulties in studying theory or solving problems that he could not solve on his own, he has the opportunity to seek oral advice. Consultations are conducted by the teacher in accordance with the established schedule.

7.5. Independent work of students

Independent work of students involves preparation for practical, lectures, tests, the performance of recommended home/individual tasks in accordance with the schedule of independent work of the work program of the discipline "Special Sections of Mathematics".

Recommendations:

- be guided by the schedule of independent work in the work program of the discipline;
- tasks must be completed in accordance with the number of the variant and drawn up in notebooks;
- to analyze unclear issues in classes and consultations;
- Preparation for the exam must be carried out on theoretical issues, the implementation and passing of tests.

7.6. Preparation for intermediate certification

Students who, for a good reason, could not score the required number of points in the current control of the module, can, in agreement with the teacher, get points before the start of intermediate certification by rewriting the tests. The main guideline in preparing for the exam is the questions given in the fund of assessment tools.

Intermediate certification is carried out on the basis of the results in accordance with the curriculum of the discipline.

It is possible to conduct intermediate certification in a remote form. In this case, all of the quizzes and exams that are listed in the discipline curriculum are completed using the Microsoft Teams platform. In this case, the exam is a rating control.

The topics of the questions and tasks of the exam in the discipline correspond to the mandatory minimum content for it. The exam ticket includes questions and tasks on sections and tasks considered during lectures, practical and laboratory classes. When preparing for the exam, it is recommended to use the sources specified in the educational and methodological support of the discipline.

The minimum number of points required for admission to the intermediate certification (exam) is 38 points (the maximum is 60 points). When summing up the results, the results of current and midterm control are taken into account, so it is important to attend classes regularly and complete all tasks on time.

A maximum of 40 points is given for the exam, a minimum of 22 points. The final score is made up of points for work during the semester, bonus points and points for the exam. The final score is converted into a grade as follows:

85-100 – "excellent", 71-84 – "good", 60-70 – "satisfactory", less than 60 – "unsatisfactory".

A student can receive up to 10 bonus points for successful participation in conferences, championships, Olympiads, competitions in mathematics, as well as for completing additional tasks of increased complexity. Points are given by lecturers only if they receive admission to the exam (from 38 points).

VIII. FUND OF ASSESSMENT TOOLS

8.1. Passport of the Appraisal Tools Fund

Nop /n	Competency Achievement Indicator	Name of the Valuation Tool
1	CPC-1.1 - Solves standard professional problems using natural science and general engineering knowledge, methods of mathematical analysis and modeling	- Tests No1-3 - Exam questions and tickets
2	CPC-1.2 - Conducts theoretical and experimental studies of objects of professional activity	
3	SC 3.1. MF-3.1 - Applies optimization techniques to the development and research of training algorithms	- Tests No1-3
4		Exam questions and tickets

8.2. Tests, tests for self-assessment of semester 3

Tests are carried out in writing in practical classes. They can be implemented in the form of options with different levels of complexity at the discretion of the teacher.

Below are examples of multi-level tests.

8.3. Semester 3 tests

Test No1. FMP. Version 1 - simplified

1. Function given $U = x \cdot \arccos(x + y) + y \cdot \sin(x + y)$. What is the expression in a point equal to? $M_0(0,0)$

$$5 - \frac{\partial^2 U}{\partial x^2} + 2 \frac{\partial^2 U}{\partial x \partial y} - \frac{\partial^2 U}{\partial y^2}?$$

2. If the $z(x, y)$ is implicitly given by the $x^2 + y^2 + 2yz - z^2 + y - 2z = 2$, then

a) partial derivatives z'_x and z'_y equal _____

b) the differential dz has the form of _____

3. The stationary point for the function $z = 2x^2 + 2y^2 - x + y - 1$ is the point _____, and the function at this point has a maximum, a minimum or no extremum?

4. Find the smallest and largest value of the function in region D:

$$z = 1 - 2x^2 + 2xy + y^2, D: \begin{cases} x \geq -1 \\ x + y \leq 1 \\ x - y \leq 1 \end{cases}$$

5. Calculate: a) $\int_{\ln 2}^{\ln 3} dy \int_4^8 ye^{\frac{xy}{4}} dx$; b) $\int_0^2 dx \int_0^x dy \int_0^{xy} x^2 yz dz$

6. The double integral for $\iint_D f(x, y) dx dy$ the region D bounded by curves: $y=x$, $y = x^2$, has the form _____

7. Changing the order of integration in the integral $\int_0^1 dx \int_{-x}^{x^2} f(x, y) dy$ leads him to the appearance of _____

8. Integral $\iint_D \sqrt{x^2 + y^2} dx dy$ by region $D: \{x^2 + y^2 \leq 9\}$ calculated in polar coordinates is: 1. 18π 2. 9π 3. 2π 4. 36π 5. 3π

9. Setting the limits of integration into $\iiint_V f(x, y, z) dx dy dz$ in the region V, bounded by surfaces: $2x+y=2$, $x+y=1$, $x=z$, $x=0$, $z=0$, leads to a multiple integral of the form: _____

10. If the body V is bounded by surfaces: $x^2 + y^2 + z^2 = 1$, $z = \frac{1}{2}$, then $z \geq \frac{1}{2}$ its volume calculated in cylindrical coordinates is equal to

$$1. \frac{11\pi}{48} \quad 2. \frac{5\pi}{6} \quad 3. \frac{11\pi}{24} \quad 4. \frac{5\pi}{24} \quad 5. \frac{5\pi}{12}$$

Test No1. FMP. Version 2

1. If $z = \arcsin(2xy + 3y)$, then a partial derivative z'_x equal _____
2. If $z = \operatorname{tg}(u^2 - 2v)$ Where is $u = 2x + \sqrt{y}$, $v = xy$, then a partial derivative z'_y equal _____
3. If the function $z(x, y)$ is implicitly given by the $x^2 + y^2 + 2yz - z^2 + y - 2z = 2$, then
 - a) partial derivatives z'_x and z'_y equal _____
 - b) the differential dz has the form of _____
 - c) the equation of the tangent plane to the surface given by this function at a point $P_0(1, 1, 0)$ has the form: _____
4. If the function $z = \operatorname{arccctg}(x - 3y)$, then a mixed derivative z''_{xy} is equal to _____
5. Find V_T ?, $V: \begin{cases} -2x + y - z + 4 = 0, \\ x = 0, y = 0, z = 0. \end{cases}$
6. Calculate: $\iint_D \frac{y^2}{x} dx dy$, $D: \begin{cases} x^2 + y^2 \leq 4x, \\ y \leq 0. \end{cases}$
7. Calculate the triple integral $\iiint_{(T)} f(x; y; z) dx dy dz$ by area (T): $f(x; y; z) = 5x + 8y$, (T): $6x - y - 2z - 3 = 0$, $x = y = z = 0$.
8. Calculate the triple integral $\iiint_{(T)} f(x; y; z) dx dy dz$, going to a cylindrical coordinate system, where: $f(x; y; z) = x(y^2 + z^2)$, (T): $x^2 + y^2 = 16z^2$, $z = 2$, $y = 0$ ($y \geq 0$).
9. Calculate the triple integral $\iiint_{(T)} f(x; y; z) dx dy dz$, switching to a spherical coordinate system: $f(x; y; z) = xy^2z$, (T): $x^2 + y^2 + z^2 \leq 4$, $z \geq 0$, $y \geq 0$
10. Calculate the curvilinear integral $\oint_{\Gamma} (2x^2 - 5y) dx + (8x - 15y^2) dy$ if D is a triangle given by the $3x - 2y = 6$, $3x + y = 2$, $x = 0$ and oriented in a positive direction.

No Rear	Assessment criteria	Maximum Points
1	Written work done in the classroom for 90 minutes (you can allow you to use your math note)	2 points
2	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	2 points
3		2 points
4		2 points
5	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	2 points
6		2 points
7	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	2 points
8		2 points
9	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	2 points
10		2 points
	Total Points	20 points

Test No1. FMP. Version 3

Execution of cases

Case 1. Substitution of variables in a double integral

Situation: Analysis of the two-dimensional distribution of model errors in domain D.

Task:

Calculate the integral $\iint_D (x + y) dx dy$, if the variables are replaced: $x = u + v$, $y = u - v$.

Case 2. Polar coordinate system for the error integral

Situation: The model has a symmetrical feature distribution around the origin.

Task:

Calculate the integral $\iint_D (x^2 + y^2) dx dy$ using the polar coordinates r, φ : $x = r \cos \varphi$, $y = r \sin \varphi$, where D is the circle of radius 1.

Case 3. Triple Integral for Data Flow Volume

Situation: the model processes three-dimensional features x, y, z , and it is necessary to estimate the total "error volume".

Task:

Calculate $\iiint_V (x + y + z) dx dy dz$ if V is a box with boundaries: $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$.

Case 4. Curvilinear integral of the 1st kind

Situation: estimation of integral error along the trajectory of change of the model feature.

Task:

Calculate $\int_C f(x, y) ds$ for the function $f(x, y) = x^2 + y^2$, if the curve C is given parametrically: $x = t$, $y = t^2$, $0 \leq t \leq 1$.

Case 5. Surface integral of the 1st kind

Situation: analysis of the model error based on the response surface of two features.

Task:

Calculate $\iint_S (x + y) dS$, where the surface S is given by the equation $z = x + y$, and $(x, y) \in [0, 1] \times [0, 1]$.

Case 6. Gradient, Divergence, and Rotor of Loss Function

Situation: analysis of a local change in the model's loss function at $P(x, y, z)$.

Problem:

Let $F(x, y, z) = (yz, xz, xy)$. Calculate:

- gradient $\varphi(x, y, z)$ for $\varphi(x, y, z) = x^2 + y^2 + z^2$,
- divergence F , $\text{div } F = \partial(F_x)/\partial x + \partial(F_y)/\partial y + \partial(F_z)/\partial z$,
- rotor F , $\text{rot } F = (\partial F_z/\partial y - \partial F_y/\partial z, \partial F_x/\partial z - \partial F_z/\partial x, \partial F_y/\partial x - \partial F_x/\partial y)$.

No back.	Assessment criteria	Maximum Points
1.	Written work can be done both individually and in teams of 2-4 people	3 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	3 points
3.	performance of more than 75% and less than 90% of the task – correct	3 points
4.	execution of tasks with a small number of errors;	3 points
5.	performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	3 points
6.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	5 points
	Total Points	20 points

Test No2. Rows. TAC
Version 1 - Simplified

- Investigate for convergence: 1) $\sum_{n=1}^{\infty} \frac{n \sqrt[n]{n+2}}{7n^2+n-1}$; 2) $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{3n^{2,5} - 2\sqrt{n} - 7n^{4/3}}{n^{3,7} + 5n^{1,2} + 9}$
- Find the convergence interval of the series: 1). $\sum_{n=1}^{\infty} \frac{2^n \cdot (x+5)^n}{\sqrt[3]{4n^2+5n}}$, 2). $\sum_{n=1}^{\infty} \frac{(x+3)^n}{(2n+3)^2 \cdot 4^n}$, 3). $\sum_{n=1}^{\infty} \frac{(x+3)^{2n}}{(2n+3)^2 \cdot 4^n}$.
- Solve the Cauchy problem for a differential equation $y' = 3(1+y^2)x^2$ with condition $y(2) = 0$.
- Solve the Cauchy problem for the equation in differentials $\frac{dx}{1-y} = \frac{dy}{1+x}$ with condition $y(1) = 2$.
- Solve the Cauchy problem for a differential equation $y'' - 5y' + 4y = 32x - 64$ with conditions $y(0) = -5$, $y'(0) = 6$.

Test No2. Rows. ODE Version2

- Investigate for convergence 1. $\sum_{n=1}^{\infty} \frac{1}{(n+2) \ln^{3/2}(n+2)}$ 2. $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$
- Find the convergence interval of the series by providing a complete study $\sum_{n=1}^{\infty} \frac{(x-3)^n}{\sqrt[3]{(2n-3)^2 \cdot 4^n}}$
- Solve the Cauchy problem: $y'' - 2y' = -e^x$, $y(0) = 3$, $y'(0) = 3$;
- Solve the Cauchy problem: $y'' + 2y' + y = x^2 + 4x + 2$, $y(0) = 1$, $y'(0) = -1$;
- Solve the system of differential equations:

$$\begin{cases} x' + x + y = 2e^{2x}, & x(0) = 0, \\ y' - 2x - 2y = -2e^{2x}, & y(0) = 1; \end{cases}$$

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow the use of your math notes)	4 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings; Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	4 points
3.		4 points
4.		4 points
5.		4 points
	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors; Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	
	Total Points	20 points

Test No2. Rows. ODE Version3

Execution of cases

Case 1. Numerical series. A prerequisite for convergence

Situation: In gradient descent, the model error decreases at each iteration, and the magnitude of the changes forms a numerical sequence. To ensure the convergence of training, it is important to understand whether the accumulated error converges.

Task:

Consider the series $\sum_{n=1}^{\infty} a_n$, where $a_n = 1 / n^p$. Determine at what values of p the series converges, which is equivalent to the condition of convergence of the sequence of error changes.

Case 2. Alternating series

Situation: when adjusting weights, neural networks with adaptive increments can change weights in opposite directions. To assess the stability of the model, it is useful to consider alternating series that simulate such oscillations.

Objective:

Investigate the series $\sum_{n=1}^{\infty} (-1)^{n+1} / n$. Determine whether the series converges and estimate the approximate value of the cumulative effect of alternating weight adjustments.

Case 3. Power series

Situation: When approximating an activation function or a loss function, models often use power series expansion to simplify analytical analysis and derivative calculations.

Task:

To decompose the function $f(x) = 1 / (1 - x)$ into a power series $\sum_{n=0}^{\infty} x^n$ and determine the convergence region.

Case 4. Fourier trigonometric series

Situation: Periodic trait data (e.g., seasonal fluctuations in demand or sensor signals) can be conveniently analyzed through Fourier series expansion to isolate frequency components.

Task:

Decompose the function $f(x) = x$, $x \in [-\pi, \pi]$, into a Fourier series: $f(x) = a_0 / 2 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$, and calculate the coefficients a_n and b_n to understand which frequencies dominate in the features.

Case 5. First-order linear differential equations

Situation: the rate of change of the model error during gradient descent can be modeled by linear remote control. Analysis of its solution makes it possible to predict the rate of convergence of the algorithm.

Task:

Solve the equation $dy/dx + y = e^x$, where $y(x)$ is the model error on the iteration of x .

Case 6. Linear homogeneous and inhomogeneous remote controls of higher orders

Situation: The dynamics of multidimensional features over time can be described by the LODU/LNDU. A homogeneous equation models the natural behavior of the system without external influences, and a heterogeneous equation models the natural behavior of the system taking into account external stimuli (for example, new data or regularization).

Task:

Solve the second-order LODU: $d^2y/dx^2 - 3 dy/dx + 2y = 0$

Solve the second-order LNDU: $d^2y/dx^2 - 3 dy/dx + 2y = e^x$ Interpret the solution as a feature change or model error over time, taking into account internal and external influences.

No back.	Assessment criteria	Maximum Points
7.	Written work can be done both individually and in teams of 2-4 people	3 points
8.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	3 points
9.	performance of more than 75% and less than 90% of the task – correct	3 points
10.	execution of tasks with a small number of errors;	3 points
11.	performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	3 points
12.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	5 points
	Total Points	20 points

Test No3. TFCP and OI. Option 1

1. To reconstruct the analytic function in the vicinity z_0 $f(z)$ of the point by the known real part $u(x, y)$ and value $f(z_0)$: $u = -3xy - 3y$, $f(0) = i$.

2. Calculate the integral along the given line: $\int_C \bar{z}|z| dz$, $C: |z| = 3$ $\text{Re } z \leq 0$ $\text{Im } z \geq 0$, . Counterclockwise traversal.

3. Decompose f-yu $f(z)$ into a Laurent series in the vicinity of the point z_0 : a) $\frac{\sin z}{z}$, $z_0 = 0$; b) $\frac{4}{z^2 + 1}$, $z_0 = i$.

4. Find all the singular points of the function $f(z)$, determine their type (order): $\frac{z-i}{(z^2+1)^3}$.

5. Using this original, find the image: $t^2 \cos 3t$.

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow the use of your math notes)	4 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	4 points
3.	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	4 points
4.	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	4 points
5.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	4 points
	Total Points	20 points

Test No3. TFCP and OI. Option 2

Execution of cases

Case 1. Development of the concept of numbers, complex numbers, geometric meaning of operations

Situation: Signal processing in machine learning often uses complex numbers to work with amplitude and phase.

Problem:

Let $z_1 = 3 + 4i$, $z_2 = 1 - 2i$. Calculate:

- $z_1 + z_2$
- $z_1 * z_2$
- z_1 / z_2
- $|z_1|$ and the $\arg(z_1)$ argument

Interpret the result as a change in the amplitude and phase of the signal.

Case 2. Function of a complex variable, a derivative of the FCP

Situation: when training neural networks with complex weights, the rate of change in the loss function for the complex variable is estimated by the derivative of the PCF.

Task:

Function $f(z) = z^2 + 2z + 1$, $z \in \mathbb{C}$. Calculate the derivative of $f(z)$ and determine the rate of change of the function at $z_0 = 1 + i$.

Case 3. Integrating complex variable functions

Situation: In the analysis of systems with a frequency response, integrals over the complex domain allow you to estimate the accumulated effect of the signal on the model.

Task:

Calculate the integral along the contour C : $\oint_C f(z) dz$, where $f(z) = 1 / (z - 2i)$, C is a circle $|z| = 3$. Interpret the result as the total contribution of the signal around the pole.

Case 4. Isolated singular points and function deduction

Situation: In the analysis of complex systems (for example, filters in neural networks or chains of linear transformations), special points of the function and their impact on the dynamics of the system are important.

Task:

Find the deduction of the function $f(z) = z / ((z - i)(z + 2i))$ at the point $z = i$. Interpret the deduction as a local influence of the features of the function on the result of signal processing.

Case 5. Laplace transform and its properties

Situation: When simulating time series or system responses to an input signal, the Laplace transform is often used to transition to the frequency domain.

Task:

Calculate $L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$, where $f(t) = e^{2t}$. Use the result to predict the response of the system at the input exponential effect.

Case 6. Applying complex analysis and Laplace to ML

Situation: When analyzing neural networks with complex weights and signal flows, it is necessary to take into account integrals, derivatives, and deductions to assess the stability of the model.

Task:

The function $f(z) = e^{zt} / (z^2 + 1)$ is given.

Find the integral $\oint_C f(z) dz$ for the loop containing the poles.

Make the Laplace transform for $f(t) = \sin(t)$ and explain the effect of the frequency component on the response of the model.

No back.	Assessment criteria	Maximum Points
3.	Written work can be done both individually and in teams of 2-4 people	3 points
4.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	3 points
5.	performance of more than 75% and less than 90% of the task – correct	3 points
6.	execution of tasks with a small number of errors;	3 points
7.	performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	3 points
8.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	5 points
	Total Points	20 points

8.4. Exam Questions and Semester 3 Tickets

Exam Questions Type 1

1. Write a formula to compute the derivative function of two variables $z = f(x, y)$ by direction $\vec{l}\{\cos \varphi; \sin \varphi\}$.
2. Write a formula to compute the derivative function of three variables $f(x, y, z)$ by direction $\vec{l}\{\cos \alpha; \cos \beta; \cos \gamma\}$.
3. What is a polar coordinate system on a plane?
4. What is a cylindrical coordinate system in space?
5. What is a spherical coordinate system in space?
6. Write a definition of a curvilinear integral of the first kind.
7. How does a curvilinear integral of the first kind reduce to a definite integral?
8. Write a definition of a curvilinear integral of the second kind.
9. Write Green's formula for a curvilinear integral of the second kind.
10. Formulate a definition of a number series and indicate the necessary condition for its convergence.
11. Formulate the signs of comparing the convergence of a numerical series.
12. Formulate d'Alembert signs of convergence of a number series.
13. Formulate the Cauchy signs and the radical Cauchy attribute of the convergence of the numerical series.
14. Formulate the integral Cauchy criterion for the convergence of a numerical series.
15. Formulate Leibniz's criterion for the convergence of a numerical series.
16. Formulate the properties of the Fourier trigonometric system.
17. Write a general view of a differential equation with separating variables.
18. What is the Cauchy problem for a differential equation?
19. Write a general view of a first-order linear differential equation.
20. What is the fundamental system of solutions of a linear homogeneous differential equation?
21. What is the characteristic equation of a linear homogeneous TAC with constant coefficients?
22. How to write the solution of a linear homogeneous TAC with constant coefficients if the characteristic equation has real, different roots?
23. How to write the solution of a linear homogeneous TAC with constant coefficients if the characteristic equation has real, multiple roots?
24. How to write down the solution of a linear homogeneous TAC with constant coefficients if the characteristic equation has complex-conjugate roots?
25. Complex variable function. The real and imaginary part of the function.
26. Cauchy-Riemann-d'Alembert-Euler conditions.
27. Integral of a function of a complex variable. Its properties.
28. Independence of the integral from the path of integration.
29. Indefinite integral of a function of a complex variable. Newton-Leibniz formula.
30. Degree series. Abel's theorem for a power series in C . Circle of convergence.
31. Laurent's series. Its properties.
32. Isolated special points. Their classification.
33. A removable singular point.
34. Polus.
35. A very special point.
36. Deduction. Relationship with the coefficients of the Laurent series.
37. Cauchy's theorem for singly connected and multiconnected domains.
38. Cauchy's integral formula.
39. Laplace transform.

Exam Questions, Type 2

40. Determination of the curvilinear integral of the first kind. Its properties. Reduction to a definite integral.
41. Definition and properties of the curvilinear integral of the second kind. Calculation of the curvilinear integral of the second kind.
42. Determination of the number series. Sum of the series. A necessary condition for the convergence of a numerical series. Properties of converging series.
43. Signs of d'Alembert convergence.
44. Cauchy convergence signs, Cauchy radical sign.
45. Alternating rows. Leibniz's sign of convergence. Absolute and conditional convergence.
46. Degree series. Abel's theorem. The region of convergence of a power series.
47. Orthogonality of a trigonometric system. Fourier series, Fourier coefficients.
48. Differential equation with separable variables.
49. A fundamental system of solutions of a linear homogeneous differential equation. Vronsky's Guide.
50. Structure of the general solution of a linear homogeneous remote control. Structure of the general solution of linear inhomogeneous remote control.
51. Linear homogeneous remote control with constant coefficients. Characteristic equation.
52. Finding a fundamental system of solutions for linear homogeneous remote control with constant coefficients.
53. Finding a partial solution of a linear inhomogeneous remote control with constant coefficients with a special right part.
54. Lagrange method of constant variation for finding a partial solution of a linear inhomogeneous PD with constant coefficients.
55. Derivative and differential of a function of a complex variable, the Cauchy–Riemann condition.
56. Basic theorem about deductions.
57. Properties of the Laplace transformation.

The exam ticket consists of two parts:

Part 1 – written work (20 points), provided that all tasks are solved correctly

Part 2 – oral answer (20 points), provided that all questions are answered correctly

Based on the results of the exam, taking into account the points received for the semester, the following grades are given:

85-100 points – "excellent" grade,

71-84 points – "good" grade,

60-70 points – "satisfactory" grade,

less than 60 points – the mark is "unsatisfactory".

8.5. Sample exam ticket (3rd semester)

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION
Federal State Autonomous Educational
Institution of higher education

"SOUTHERN FEDERAL UNIVERSITY"

EXAM TICKET No 0

By discipline **SPECIAL SECTIONS OF MATHEMATICS, semester 3**

Structural unit. Institute of Computer Technologies and Information Security

1. Formulate a theorem on mixed derivatives of FNP.
2. A fundamental system of solutions of a linear homogeneous differential equation. Vronsky's Guide.
3. For the $f(x; y) = \sqrt{5x^2 + 5y^2}$ Calculate Double Integral $\iint_D f(x, y) dx dy$ by region D , lying in the third quarter and bounded by the lines $x^2 + y^2 = 4$, $y = 0$, $y = x \cdot \sqrt{5}$.
4. Calculate the ternary integral $\iiint_{(T)} f(x; y; z) dx dy dz$ by going to a cylindrical coordinate system, where (T) is the body bounded by the specified surfaces: $f(x; y; z) = xy + z$, (T): $z = 9(x^2 + y^2)$, $z = 2$
5. Calculate the curvilinear integral $\oint_{\Gamma} (x^2 - 7y) dx + (4x - 5y^2) dy$ if G is a triangle given by the lines $3x - 2y = 5$, $3x + y = 2$ and $x = 0$ oriented in the positive direction.
6. Investigate the convergence of a numerical series $\sum_{n=1}^{+\infty} (-1)^n \cdot \frac{2n^{2/7} - 5\sqrt{n} + 6n^{3/2}}{9n^{2/7} + 7n^{1/6} + 2}$.
7. Solve the Cauchy problem for a differential equation $y''' - 6y'' + 8y' = 16x - 52$ with terms and conditions $y(0) = 1$, $y'(0) = 22$.
8. Using this original, find the image: $e^{-4t} \sin 3t \cos 2t$.

Compiled by: _____ A.G. Klovo
_____ I.A. Bugaeva
Head. Department. FMIO _____ G.V. Kupovykh
« » 202..

Evaluation criteria:

35 - 40 points are given to a student if a detailed answer to theoretical questions is given with an appropriate definition, accompanying theorems and consequences, as well as a practical task is correctly solved;

30 - 34 points are given to a student if a correct, but not complete answer to the question to theoretical questions is given, and the practical task is also correctly solved;

20 – 29 points are given to a student when he has answered no more than one theoretical question and has partially coped with the solution of the practical task;

Less than 20 points are given in case of an incorrect answer to both theoretical questions.